

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ERROR CORRECTING CODES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1.
 - (a) Justify the statement "All prefix codes are uniquely decodable codes but not vice versa" with an example. 3(CO1)
 - (b) What is meant by LRC and VRC with respect to error detecting codes? Give example of the same. 3(CO1)
 - (c) A DMS has six symbols $x_1, x_2, x_3, x_4, x_5, x_6$ with
 $P(x_1) = 0.35$ $P(x_2) = 0.25$ $P(x_3) = 0.2$ $P(x_4) = 0.1$ $P(x_5) = 0.06$ $P(x_6) = 0.04$
 - (i) Construct a Shannon-Fano code for X. Calculate efficiency of the code.
 - (ii) Repeat for Huffman code and compare the results. 4(CO1)
 2.
 - (a) What are Hamming codes ? Write properties of Hamming Codes. 3(CO2,3)
 - (b) Compare Go Back N ARQ and Selective Repeat ARQ Systems. 3(CO2,3)
 - (c) Elaborate synchrode decoding process in linear block codes. 4(CO1)

3. A message 10 11 01 is to be transmitted in cyclic code with a Generator polynomial $G(p) = p^4 + p^3 + 1$.
Obtain the transmitted code word. How many check bits does the encoded message contain ? Draw the encoding arrangement for the same. 10(CO2,3)
4. A Convolution Code is described by $V_1 = [1\ 1\ 1\ 1]$; $V_2 = [1\ 0\ 1]$;
Draw the convolution encoder.
The received sequence for above encoder 010010111
Decode the sequence using Viterbi decoding Algorithm.
Comment on Metric Divergence Effect. 10(CO5)
5. Differentiate between Recursive and Non Recursive Codes. Illustrate Recursive encoder with a suitable example. 10(CO4)
6. (a) Discuss encoding and decoding process of Turbo codes with one example. 5(CO4)
(b) Elaborate various types of Interleaver Design in Turbo codes. 5(CO4)

